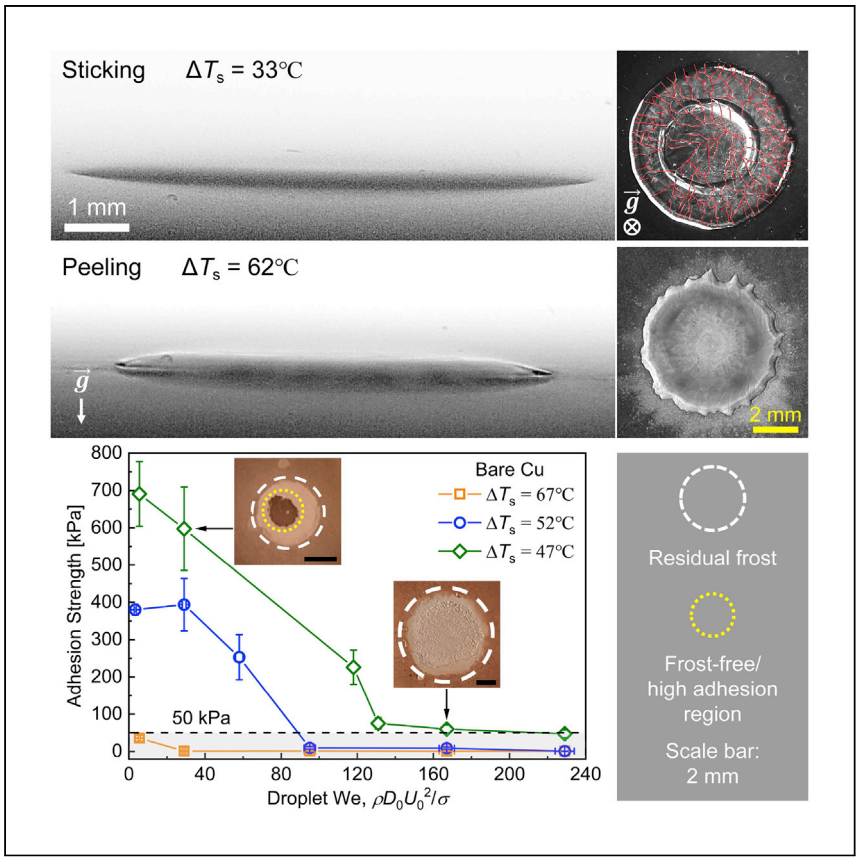
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Ultra-low ice-substrate adhesion and self-deicing during droplet impact freezing



A droplet impacting on a cold surface usually strongly sticks to the substrate during solidification. Jin et al. quantify the ultra-low ice-substrate adhesion and achieve spontaneous deicing on droplet impact freezing at high-surface supercoolings (>30°C).

Puhang Jin, Xiao Yan, Muhammad Jahidul Hoque, ..., Saijie Cai, Wenquan Tao, Nenad Miljkovic

yan-x13@tsinghua.org.cn (X.Y.)
wqtao@mail.xjtu.edu.cn (W.T.)
nmiljkov@illinois.edu (N.M.)

Highlights

Clarification of the relationship among frost growth, peeling, cracking, and adhesion

Characterization of heterogeneous adhesion of droplet impact icing to substrates

Development of a regime map for self-peeling and ultra-low adhesion (0–50 kPa)

Demonstration of passive self-deicing for single and multiple impacting droplets

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Ultra-low ice-substrate adhesion and self-deicing during droplet impact freezing

Puhang Jin,[1,2,6] Xiao Yan,[1,6,*] Muhammad Jahidul Hoque,[1] Kazi Fazle Rabbi,[1] Soumyadip Sett,[1] Jingcheng Ma,[1] Jiaqi Li,[1] Xiaolong Fang,[1] James Carpenter,[1] Saijie Cai,[2] Wenquan Tao,[2,*] and Nenad Miljkovic[1,3,4,5,7,*]

SUMMARY

A water droplet impacting onto a supercooled surface is typically considered to freeze and adhere to the substrate. This ice accretion poses safety and economic threats to transportation infrastructure, power generation/transmission systems, and telecommunication facilities. Here we report the observation of ultra-low ice-substrate adhesion (0–50 kPa) and remarkable self-deicing during droplet-impact freezing on copper surfaces having medium to high supercooling (30°C–80°C). Mechano-thermo-hydraulic coupling during droplet-impact freezing governs the ice-substrate adhesion by gapping the droplet-substrate contact, enabling self-peeling facilitated by thermal-mechanical stress relaxation. We observe a strong adhesion region in the center of the frozen droplet, which determines the adhesion strength, and develop a regime map to delineate the dependence of adhesion/peeling on droplet inertia, substrate supercooling, and surface wettability. Our work demonstrates key mechanisms governing ice-substrate adhesion during impact icing and presents an approach to passive self-deicing.

INTRODUCTION

High supercooling above 30°C is commonly encountered in polar regions and in industries processes such as aviation, biological tissue preservation, long-distance cold-chain transportation, and next-generation refrigeration.[1–6] Ice accretion on supercooled surfaces is ubiquitous and presents significant safety and economic concerns.[1,2] Ice can accumulate on a surface by desublimation, freezing after condensation, or freezing after droplet/substrate collision.[3] All commercially used deicing systems are either energy or resource intensive.[4] A more desirable approach to mitigate icing is to lower ice-substrate adhesion or even prevent ice formation completely rather than having to address its buildup. In this regard, an increasing body of work has focused on developing "icephobic" surfaces.[2–6]

Superhydrophobic surfaces (SHPs) have been shown to possess remarkable anti-icing capacity, including freezing delay, inhibition of condensation and frosting, repulsion of impacting water droplets, and reduction of ice adhesion.[7–11] Icephobicity has been attributed to the numerous air pockets sustained within the micro-/nano-structures that minimize contact area and reduce surface energy. However, micro-structures, which may induce mechanical interlocking because of frost nucleation within the textures,[12] deteriorate the anti-icing performance of SHP in humid environments.[9,13,14] To address this challenge, researchers have developed slippery liquid-infused porous surfaces (SLIPSs) to create defect-free slippery interfaces and

[1]Department of Mechanical Science and Engineering, University of Illinois at Urbana–Champaign, Urbana, IL 61801, USA

[2]Key Laboratory of Thermo-Fluid Science and Engineering of Ministry of Education, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, P. R. China

[3]Department of Electrical and Computer Engineering, University of Illinois at Urbana–Champaign, Urbana, IL 61801, USA

[4]Materials Research Laboratory, University of Illinois, Urbana, IL 61801, USA

[5]International Institute for Carbon Neutral Energy Research (WPI-I2CNER), Kyushu University, 744 Motooka, Nishi-ku, Fukuoka 819-0395, Japan

[6]These authors contributed equally

[7]Lead contact

*Correspondence: yan-x13@tsinghua.org.cn (X.Y.), wqtao@mail.xjtu.edu.cn (W.T.), nmiljkov@illinois.edu (N.M.)

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minimize contact angle hysteresis.[15–18] However, icephobicity of both SHPs and SLIPSs is short-lived due to difficulty in sustaining the surface structures or lubricating oil during icing/deicing cycles.[18] In addition, as surface supercooling increases (>30°C), ice formation by impacting droplets becomes unavoidable even on the best icephobic surfaces,[19,20] making anti-icing by repulsion of impacting droplets impractical. Therefore, the fundamental understanding of icing dynamics and exploration of new passive deicing strategies have been central to the design and operational safety of many ice-suffered scenarios.

Droplet impact on supercooled substrates governs the mechanism of ice formation for many situations.[21,22] At large surface supercooling, the freezing timescale is shorter than the droplet spreading time, meaning the icing-governing hydrodynamic interactions between the droplet, surrounding gas, and solid substrate remain poorly understood.[23–25] For tin and hexadecane droplets,[26,27] self-delamination can occur between the solidified droplets and the impacted supercooled surfaces. However, their peeling dynamics and mechanisms are quite different.[26,27] Although more recent work has demonstrated self-peeling of impact icing water droplet on supercooled substrates,[21] whether ultra-low adhesion and self-deicing work for water (ice) and its manipulation strategies remain ill-defined because of the great differences in their thermal and mechanical properties and the lack of quantifying droplet-substrate adhesion strength.

Here we report the ultra-low adhesion (0–50 kPa) and remarkable self-deicing (ice falling by its own weight, splitting into debris, or detached by neighboring droplets) during water droplet impact freezing on copper surfaces for a range of substrate supercooling (from 30°C to 80°C). We characterize the impact of icing dynamics and adhesion strength behavior using high-speed imaging and *in situ* shear strength testing. We demonstrate that ultra-low adhesion is dominated by the interactions between the impacting droplet and entrapped air during instantaneous freezing at the droplet-substrate interface. The low adhesion is further facilitated by the subsequent thermal-mechanical stress relaxation dynamics, which bend the ice flake rather than generating cracks.[28] Regime maps are created to delineate the dependence of adhesion strength and self-peeling on substrate supercooling, surface wettability, and droplet inertia. Our work not only provides fundamental insights into the mechanisms of droplet impact freezing but also demonstrates an avenue of utilizing fast-freezing dynamics, thermal contraction stress, aerophilic surface, and high thermal conductivity materials to achieve ultra-low apparent ice adhesion and even passive self-shedding of ice lamella.

RESULTS

Sticking or self-peeling

By increasing the substrate supercooling (Figures 1A and 1B), we observed that the circumferential edge of the frozen droplet detached from the substrate at 33 ms < t < 106 ms (Figure 1B) rather than stuck to the impacted surface (Figure 1A), leaving a distinct air gap between the ice and impacted surface with a maximum thickness, defined herein as the peeling height (δ), of $\delta = 59 \pm 9$ μm at the radial edge (see Video S1). To understand the mechanisms governing peeling, the peeling height (δ) as a function of time (t) and the bending curvature (K) of ice lamella as a function of substrate supercooling (ΔT_s) were extracted from high-speed videos. Peeling height, δ (Figure 1C), and bending curvature, K (Figure 1D), are zero when ice sticks to the impacted surface. Peeling was initiated immediately after completion of bulk freezing on the bare Cu surface, with peeling initiation coinciding with

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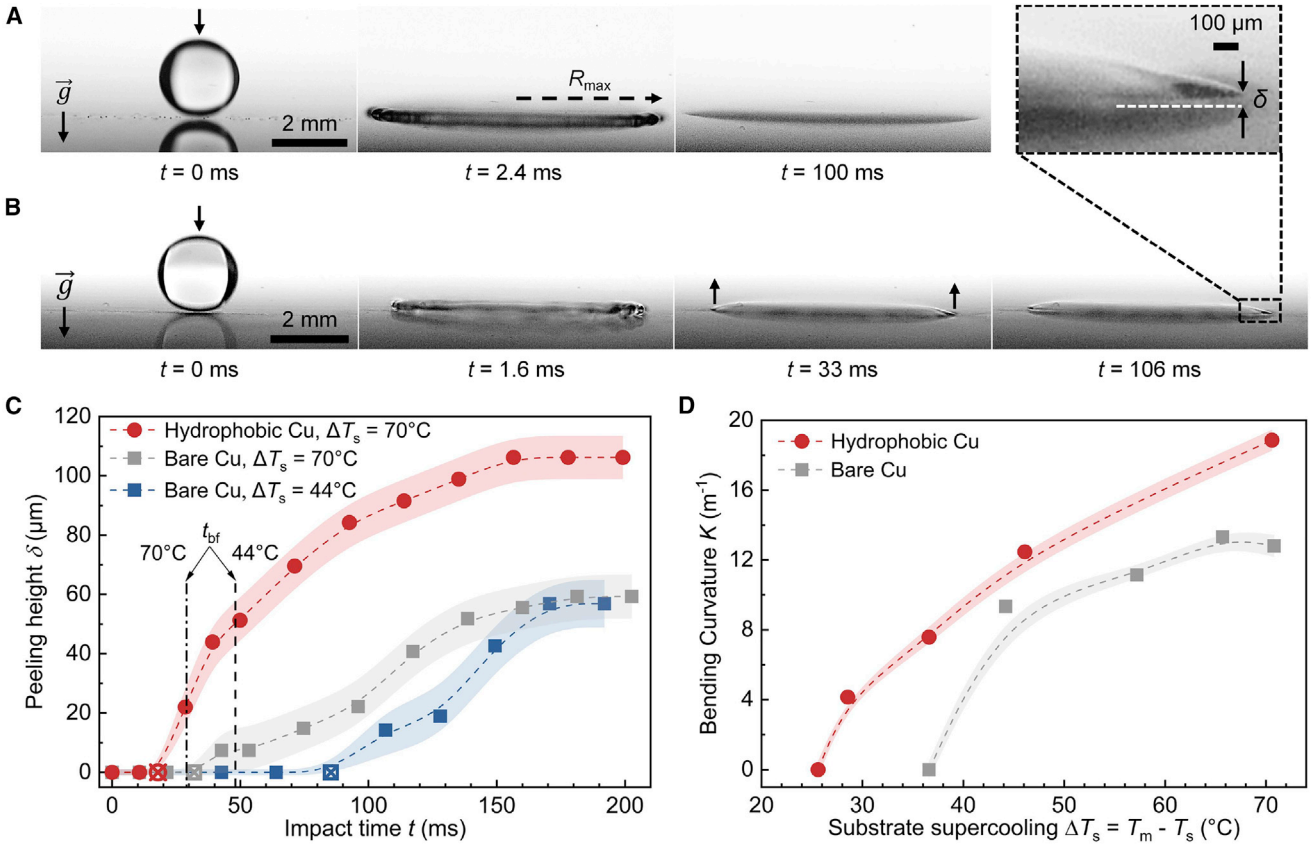


Figure 1. Droplet impact and freezing dynamics on supercooled surface

(A) Side-view high-speed time-lapse images showing water droplet impact and freezing dynamics on a bare mirror-finished Cu surface at substrate supercooling of ΔT_s = 18°C. Ice stuck to the impacted surface.

(B) Side-view high-speed time-lapse images showing water droplet impact and freezing dynamics on a bare mirror-finished Cu surface at substrate supercooling of ΔT_s = 62°C (see Video S1). Self-peeling was observed with increased substrate supercooling.

(C) Peeling height (δ) as a function of impact time (t) for the bare Cu and hydrophobic Cu surfaces. The two vertical lines represent the theoretical bulk freezing moment t_{bf} (see Note S1) when ΔT_s = 44°C (dashed line) and ΔT_s = 70°C (dashed-dotted line).

(D) Bending curvature (K) of the ice lamella as a function of supercooling (ΔT_s) on the bare Cu and hydrophobic Cu surfaces. Test conditions: D_0 = 2 ± 0.05 mm, T_d = 23 ± 0.5°C, and We = 167 (U_0 = 2.45 m/s).

the bulk freezing time, t_{bf} (see Figure S1; Note S1), at very large supercooling (ΔT_s = 70°C). At moderate supercooling (ΔT_s = 44°C, blue curve in Figure 1C), peeling initiated at $t \approx$ 85 ms (marked by the blue square symbol with a cross) and ended at $t \approx$ 170 ms, achieving a peeling height of δ = 60 ± 8 μm. At large supercooling (ΔT_s = 70°C, gray curve in Figure 1C), peeling initiated earlier at approximately $t \approx$ 33 ms, with ending time and peeling height δ approximately equivalent to those observed at moderate supercooling (ΔT_s = 44°C). The bending curvature (K) initially showed an increasing trend (from 0 to approximately 12.5 m^{-1}) with substrate supercooling increased from 37°C to 66°C for the bare Cu (gray curve, Figure 1D). With a further increase in surface supercooling (>66°C), K remained relatively constant.

In addition, our experiments conclusively quantify the role of entrapped air in peeling dynamics by changing surface wettability from hydrophilic to hydrophobic. On the hydrophobic Cu surface (red curve, Figure 1C), the ice detached from the substrate at $t \approx$ 17 ms (see Figure S2), much earlier than the time required to complete bulk freezing ($t \approx$ 29 ms, black dashed-dotted line in Figure 1C). The peeling height of δ = 105 ± 8 μm was 75% larger than its bare Cu counterpart (60 ± 8 μm). In

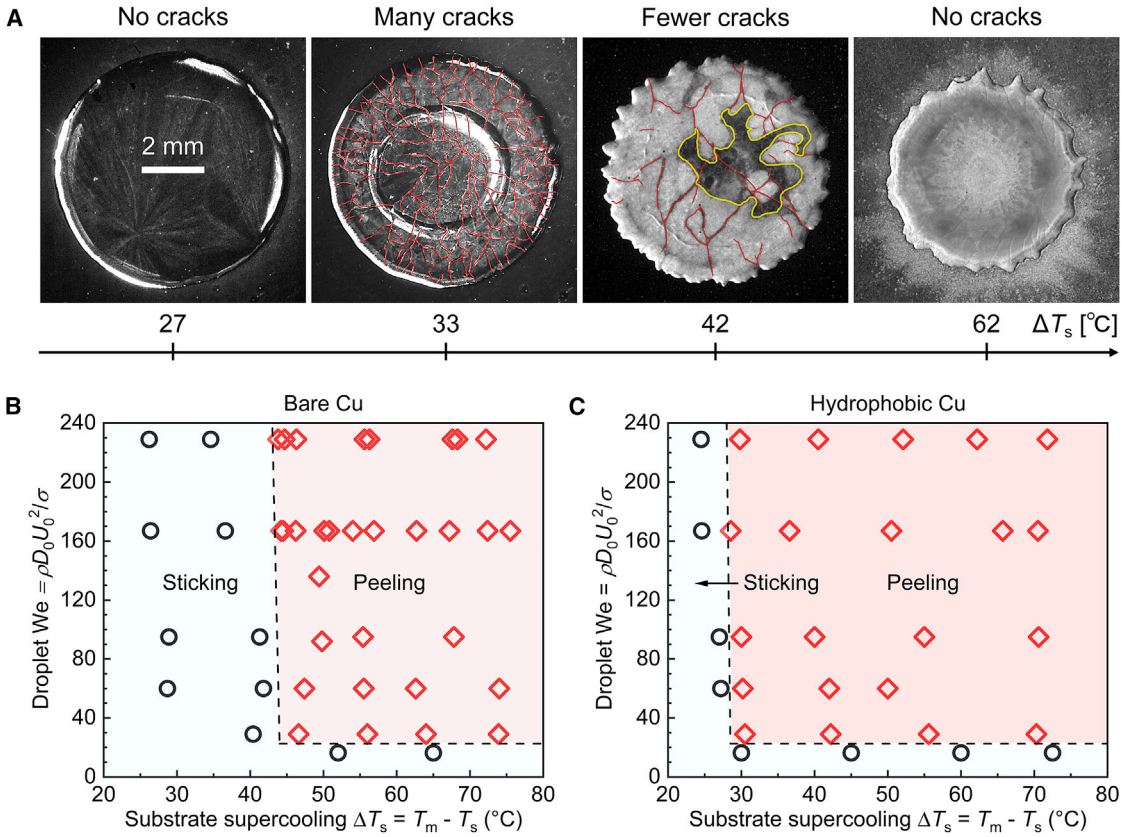


Figure 2. Ice cracking and regime map of peeling/sticking

(A) False-colored top-view images showing the evolution of crack patterns generated after droplet impact icing on bare Cu. (A) was tested with $D_0 = 2 \pm 0.05$ mm, $T_d = 23 \pm 0.5$°C, and We = 167 ($U_0 = 2.45$ m/s).

(B) Regime map of peeling versus sticking during droplet impact freezing for bare Cu surface. The partitions delineating each regime are presented by the vertical and horizontal lines, with the transitional substrate supercooling $\Delta T_s \approx 43$°C and We ≈ 20.

(C) Regime map of peeling versus sticking during droplet impact freezing for hydrophobic Cu surface. The partitions delineating each regime are presented by the vertical and horizontal lines, with the transitional substrate supercooling $\Delta T_s \approx 29$°C and We ≈ 20. The light blue region (black circles) represents the sticking regime, while the light red region represents the peeling regime (red rhombi). (B) and (C) were tested with $D_0 = 2 \pm 0.05$ mm and $T_d = 23 \pm 0.5$°C. Error bars are smaller than the data points and are not shown for clarity.

addition, the hydrophobic coating gives rise to additional bending curvature (18.9 m^{-1}) and lowers the transitional supercooling of self-peeling to approximately 29°C (red curve, Figure 1D). However, as indicated by Song et al.,[21] the deflection of ice on functionalized hydrophobic silicon is much smaller than on bare silicon.

Mechano-thermo-hydraulic coupling

For the optically non-transparent surfaces studied here, it is not possible to study interference fringes from the bottom to show the dynamics of entrapped air at the droplet-substrate interface,[26] whereas top-view time-lapse image analysis (Figure 2A), enabled by the transparency of ice, offers additional information to identify the peeling governing mechanisms (see Note S2). At low supercooling ($\Delta T_s \leq 27$°C), we observe no cracks within the ice.[28] At a higher supercooling ($\Delta T_s = 33$°C), numerous cracks form within the ice lamella (see Figure S3A; Video S2). Remarkably, the number of cracks decreases with further increase in supercooling ($\Delta T_s > 33$°C), showing almost no cracks at a very large supercooling ($\Delta T_s = 62$°C) (see Figure S3B). We note that the side-view images of Figure 1B are for the same experimental case as shown in Figure 2A at $\Delta T_s = 62$°C.

Ice is subjected to rapid thermal contraction when the substrate temperature (T_s) is reduced to below the freezing temperature (0°C for water).[29] At low supercooling ($\Delta T_s \leq 27°C$), thermal contraction stresses are too small to generate new interfaces (cracks). Larger supercooling offers additional thermal contraction energy for crack formation.[28–30] We hypothesize that a further increase in ΔT_s (>33°C) prompts the instant freezing at the droplet-Cu interface, leaving many air gaps between the thin ice layer and impacted surface within the course of droplet spreading. The observed frosting on the substrate underneath the ice layer, indicated by the bright color in contrast with copper background (see Figure 2A), supports our hypothesis. When the surface supercooling is small ($\Delta T_s = 27°C$), the ice lamella is optically transparent, as manifested by the contiguous brightness between the ice base and the surrounding region.[28,31–33] However, at larger supercooling ($\Delta T_s = 42°C$), the brightness within the impact area becomes heterogeneous. The majority of the ice region is white (not as dark as the surrounding area), indicating that frost has formed between the ice and the supercooled surface during freezing front propagation, except for the area encircled by the yellow curve in Figure 2A where brightness is lower. Thus, the ice flake does not have intimate contact with the impacted surface at this high supercooling, with only the yellow curve encircled area ($\Delta T_s = 42°C$ in Figure 2A) having relatively good contact. Considering that the surrounding area has less frost than that beneath the ice flake ($\Delta T_s = 42°C$ in Figure 2A), we demonstrate that water vapor, caused by the latent heat released during water film solidification, desublimates into frost (see Figures S3B and S3C).[34] Hence we can say that the thermo-mechanical stress induced by thermal contraction bends the ice flake from the substrate instead of generating cracks when ice adheres loosely with the substrate at high-substrate supercooling. The low adhesion between ice and substrate, induced by instantaneous freezing at the droplet-substrate interface on contact, is a premise of self-peeling.

The nucleation and freezing time of the thin water layer near the supercooled interface during droplet spreading are governed by the temperature at the solid-liquid interface T_{int}. We approximate the interface temperature, T_{int}, as the contact temperature between two semi-infinite bodies (droplet and substrate) that are bought into contact:[32]

$$T_{int} = T_s + \frac{T_d - T_s}{1 + e_s/e_d}, \qquad \text{(Equation 1)}$$

where $e = \sqrt{k\rho c_p}$ is the thermal effusivity (see Table S1), and subscripts s and d denote substrate and droplet, respectively. We verified the role of T_{int} in governing the ice-substrate adhesion by changing substrate thermal effusivity, e_s. We can see that stainless steel (SS) exhibits close transitional interface temperature T_{int} (~41°C) of sticking or peeling to that of Cu, despite the differences in their transitional T_s (see Figure S4; Note S3).

The rapid cooling-induced thermal contraction can be quantified using an energy analysis.[35,36] By integrating the elastic energy density (see Note S1) within the frozen ice layer, with the ice layer thickness $h(t_p)$ at t_p, with the corresponding temperature field (see Figure S1), we have the total thermal contraction energy (E_{tc}) from the contact moment to t_p as

$$E_{tc} = \frac{E\alpha^2}{2} \Delta T_s^2 \pi R_{max}^2 h(t_p), \qquad \text{(Equation 2)}$$

where $E = 9.33$ GPa is the ice Young's modulus, and $\alpha = 5.3 \times 10^{-5}\ K^{-1}$ is the thermal expansion coefficient of ice.[37]

The dissipation of thermal contraction energy can be divided into three components: detachment of the ice from the impacted surface, formation of cracks, and global plastic deformation (bending). Detachment of the ice flake consumes negligible energy when compared with the total thermal contraction energy E_{tc} (see Note S4). The energy consumed by crack formation having a total length C and a nominal width $h_0 = D_0^3/6R_{max}^2$ can be roughly estimated by

$$E_c = 2G_c C h_0, \quad \text{(Equation 3)}$$

where $G_c \approx 1$ kg/s^2 is an associated energy per unit surface when generating a new interface, known as the Griffith energy.[38] With ΔT_s increased from 33°C to 42°C (Figure 2A), the ratio between E_c and E_{tc} rapidly decreases from 46% to 9.8% (see Figure S5). Hence the plastic deformation energy governs the intensity of self-peeling, and more energy budget for ice bending is available at larger supercooling.

The observed peeling dynamics on hydrophobic Cu (Figures 1C and 1D) can be explained by the above energy analysis and competing relationship between cracking and peeling. More air can be entrapped by the impacting droplet with increased surface hydrophobicity; prolonging the interaction time between entrapped air, freezing liquid, and substrate; and inducing additional air-gap initiators at the droplet-solid interface that eventually reduce adhesion. The lower adhesion leads to a smaller chance of crack formation (see Figure S6) and less energy consumption by cracks, E_c, which eventually results in a greater thermal contraction energy budget for bending. Thus, as long as the thermal contraction energy exceeds the requirement to achieve deformation/bending, peeling can be triggered for partially solidified droplets before bulk freezing despite that most bending happens during thermal homogenization after bulk freezing.[26,28]

Droplet inertia, characterized by the droplet Webber number (We), alters self-peeling by influencing the droplet spreading dynamics, and hence the ice morphology. Peeling is prohibited by increased ice thickness (see Figure S7). We develop a regime map of self-peeling for both bare (Figure 2B) and hydrophobic Cu (Figure 2C) surfaces. The partitions can be represented by a quasi-vertical line and a horizontal line, with the transitional substrate supercooling $\Delta T_s \approx 43$°C and 29°C for bare and hydrophobic Cu surfaces, respectively, and We $\approx$ 20 for both. It is noted that the transitional supercooling of peeling or sticking for hydrophobic Cu is close to the supercooling (27°C) below which no cracks can be generated, and that for bare Cu is very close to the transitional temperature (42°C) of crack pattern transformation from fragmentation to hierarchical cracking, which forms when the first crack is generated just at the moment of bulk freezing, observed in a prior study.[28] These two coincidences demonstrate evidence of the direct competing relationship between peeling and cracking for brittle materials after droplet impact solidification.

Adhesion regime map

Although global deformation is governed by thermal contraction energy relaxation, further reduction of ice adhesion, apart from the fast-freezing-induced intrinsic low adhesion between the ice and substrate at large supercooling, can be achieved by partially detaching the ice from the supercooled surface. We characterize the adhesion strength for impact droplet freezing using an *in situ* ice soldering method (Figure 3A) coupled with shear strength measurements.[39] We divided the shear force by the vertical projected area of the ice lamina when calculating the adhesion strength, resulting in an apparent shear adhesion strength. The apparent shear adhesion is widely used to characterize the ice adhesion and to guide the design of ice-phobic coatings.[2,12,16,39]

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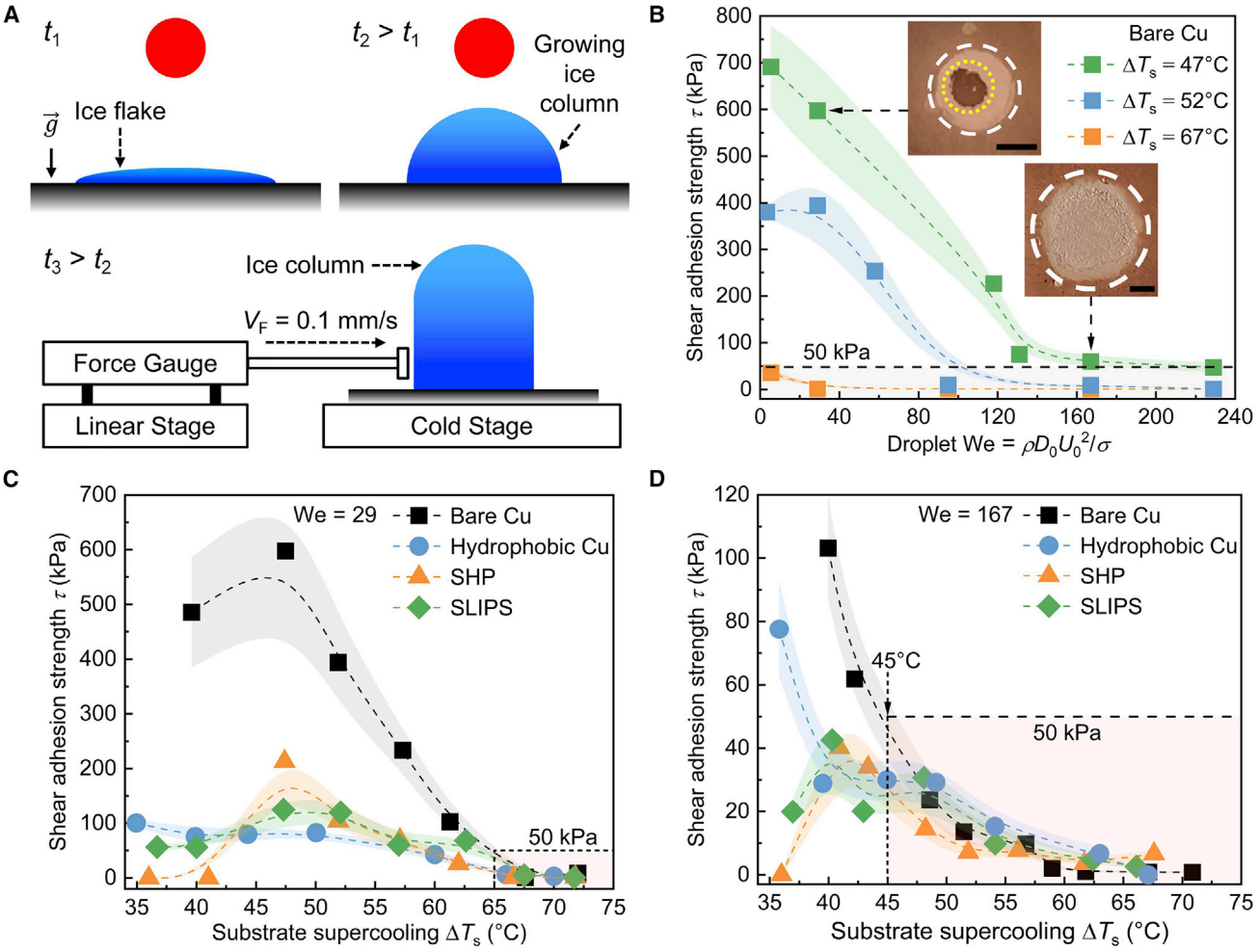


Figure 3. Apparent adhesion strength of ice formed by droplet impact freezing

(A) Schematic diagram showing the shear strength test setup. An ice column is formed by sequential droplet impact onto the supercooled surface and then pushed by the tip of a force gauge to measure the apparent adhesion shear strength. Schematics not to scale.

(B) Apparent ice shear adhesion strength τ as a function of droplet We on bare Cu. Scale bars of insets represent 2 mm.

(C) Apparent ice shear adhesion strength τ as a function of supercooling (ΔT_s) and surface properties (materials) for We = 29.

(D) Apparent ice shear adhesion strength τ as a function of supercooling (ΔT_s) and surface properties (materials) for We = 167. The 0-kPa adhesion strength of the superhydrophobic CuO surface (SHP) in (C) and (D) denotes droplet repulsion and rebound on impact with the supercooled surface. Tests of (B)–(D) were conducted with T_d = 23°C ± 0.5°C and D_0 = 2 ± 0.05 mm. Shaded areas in (B)–(D) represent the measurement error of apparent ice shear adhesion strength τ.

The apparent shear adhesion strength, τ, shows a sharp and then a mild declining trend with increasing droplet We when ΔT_s = 47°C and 52°C (Figure 3B) for bare Cu. Eventually, the adhesion strength exhibits a very low value (< 50 kPa) at large droplet We (We > 90 for ΔT_s = 52°C and We > 200 for ΔT_s = 47°C). At larger surface supercooling (ΔT_s = 67°C; Figure 3B), the low adhesion limit (<50 kPa) was established within the entire range of droplet We tested.

We identify a heterogeneous adhesion behavior, manifested by frost residuals after conducting the shear strength test, during droplet impact icing. Frost layers (marked by the white dashed circles in the two insets in Figure 3B) were observed on the impacted area in both low and high adhesion cases, in accordance with top-view observations of frost formation in Figures 2A and S3B. The difference between these cases is that at decreased droplet We (ΔT_s = 47 °C, We = 29), an area with less frost

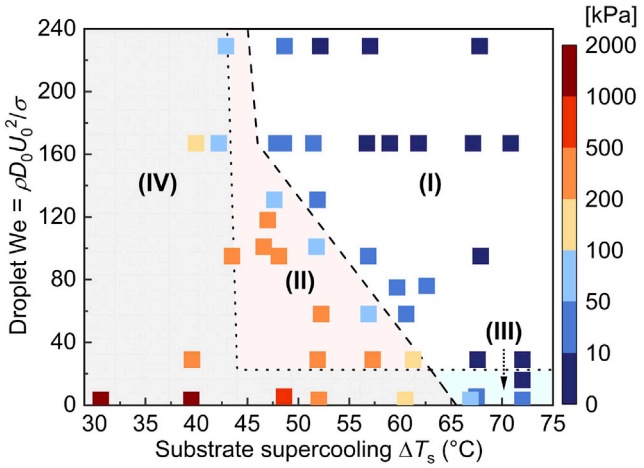


Figure 4. Ice-substrate apparent adhesion strength regime map

The regime map is delineated by the low adhesion curve (dashed line) and peeling curve (dotted line) into four regimes: (I) peeling with low adhesion, (II) peeling with high adhesion, (III) sticking with low adhesion, and (IV) sticking with high adhesion. All test data were obtained for bare Cu substrates with droplet diameter D_0 = 2 ± 0.05 mm and droplet temperature T_d = 23°C ± 0.5°C.

(the Cu is exposed) exists at the central region of the impacted area (marked by the yellow dotted circle in the left inset of Figure 3B). This observation indicates that the central region has good contact with the substrate, hence vapor, induced by latent heat release, cannot penetrate into the droplet-substrate interface to form frost during freezing front propagation. The central Cu-exposed region, i.e., strong adhesion area, along with the adhesion strength, reduces with increased ΔT_s and droplet We (see Figure S8; Note S5).

The adhesion strength decreases with increased supercooling for all tested surfaces at both droplet We values (Figures 3C and 3D) despite the fact that the functionalized surfaces (hydrophobic Cu, SHP, and SLIPS) are less sensitive to increased supercooling. An adhesion strength of zero for the SHP surface represents partial or full repulsion (see Video S3) of impacting droplets. Bare Cu showed larger τ when compared with the other surfaces for low supercooling, with the differences decreasing with increased supercooling. Once a threshold supercooling T_{thr} is reached, the bare Cu exhibited comparable low adhesion strength with the other three icephobic surfaces, with T_{thr} approaching 65°C and 45°C for We = 29 and 167, respectively. Low adhesion strength (0–50 kPa) is expected for other industrial metals at different threshold supercooling.

An adhesion strength regime map for bare Cu (Figure 4) was developed by defining that adhesion smaller than 50 kPa represents the ultra-low-adhesion regime.[39] Applying partitions from the peeling regime map (Figure 2B) to Figure 4, we create four distinct regimes. Regimes I–IV denote peeling with low adhesion, peeling with high adhesion, sticking with low adhesion, and sticking with high adhesion, respectively. Peeling and ultra-low-adhesion regimes coincide in regime I (large ΔT_s and We), while sticking and high adhesion meet in regime IV (small ΔT_s and We). Passive self-peeling can occur for ice having τ as high as 400 kPa in regime II. At low droplet We and high substrate supercooling (ΔT_s, regime III), peeling is suppressed by large ice thickness. However, ultra-low adhesion is maintained at small We (<20) as long as the substrate supercooling is high enough.

Passive self-deicing

Impact freezing can be greatly mitigated via passive and spontaneous self-deicing by rationally increasing droplet inertia, substrate supercooling, and aerophilic properties to optimize the impact-freezing interactions. To test our hypothesis, we conducted additional experiments with larger droplets and impact velocities on a hydrophobic Cu surface. Two previously unidentified self-deicing mechanisms via thermal contraction energy dissipation are identified. We find that ice lamella can readily slide from the hydrophobic Cu surface by its own weight when tilting the surface (Figure 5A; see Video S4). Further, increasing ΔT_s to 75°C with D_0 = 3.7 mm results in the ice lamella suddenly splitting into several fragments approximately 1.6 s after bulk freezing (Figure 5B; Video S5). The two observed mechanisms of self-deicing have not been observed on bare Cu, even for supercooling as high as 85°C. Given the energy conversion process, we expect ice splitting to occur on bare hydrophilic Cu surfaces at a lower T_s, which is not tested here because of the cooling capacity limits of our experimental setup. Our findings paint a more complex picture of thermal contraction energy dissipation. When ice adheres to the substrate tightly, cracks are the primary method for thermal stress dissipation. For low ice adhesion (large ΔT_{int}), global deformation (bending) governs energy release. As the substrate supercooling increases further, ice splitting, coupled with peeling, accounts for energy release.

To quantify the applicability of low adhesion and thermal stress-induced passive self-deicing, we also visualized droplet impact icing of multiple sequential droplets. We observed that the latter droplets, overlapping with neighboring droplets, induced directional thermal-mechanical forces toward the subsequent droplets and fully detached the prior frozen droplet from the substrate (Figure 5C). In addition, as another demonstration of multiple droplet impact icing (see Video S6), we sprayed water columns using a rinsing laboratory bottle onto the supercooled surface (ΔT_s = 85°C). When the water column impacts the top-middle of the cold surface, it formed an ice flake with a spindle-like shape. The spindle tip detached and tilted outward from the cold surface. Immediately after, an ice flake at the left-top corner of the cold surface broke and fell from the supercooled surface. This phenomenon demonstrates the potential application of manipulating the adhesion/peeling behavior of droplet swarms by employing the thermal-mechanical stress.

DISCUSSION

The entrapped air beneath the impacting droplet accounts for the discontinuous contact between the freezing droplet and the substrate.[21,26] Thus, quantitative characterization of the entrapped air dynamics on supercooled surfaces, especially considering the effect of substrate wettability, needs to be conducted to better understand the unprecedented ultra-low-adhesion regime. In the future, it would be interesting to incorporate rationally designed close-cell brick or honeycomb microstructures on the impacted surface to enable better entrapment of air,[19] which is likely to further promote self-peeling and adhesion reduction. In addition, the ultra-low adhesion is observed with quasi-static water droplets (H = 5 mm) on highly supercooled (ΔT_s = 68°C) bare Cu surfaces in spite of sticking (no peeling). We speculate that the separation of dissolved air at the droplet-substrate freezing interface also plays an important role in reducing the adhesion strength.[40]

Self-peeling was not observed on icephobic surfaces studied here (SHPs and SLIPSs), although they have low adhesion with ice. The main reason is the low thermal conductivity of the CuO nanoblade structures (see Figure S9) and the infused oil, which

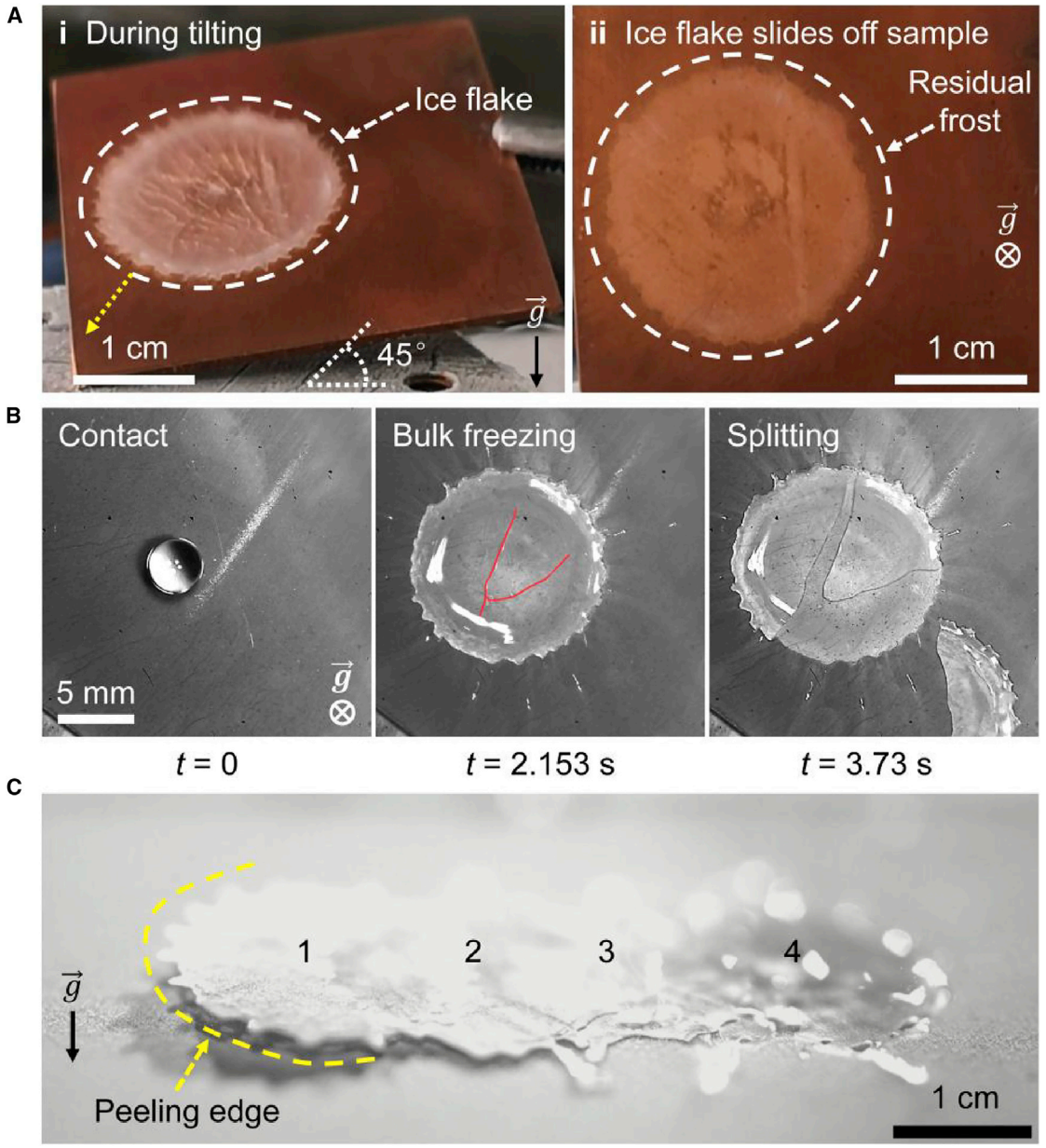


Figure 5. Spontaneous deicing of single and multiple droplets

(A) Photographs of self-deicing during droplet impact freezing on a hydrophobic Cu surface (see Video S3) with ΔT_s = 67°C, D_0 = 5.3 ± 0.05 mm, T_d = 23 ± 0.5°C, and We = 642 (U_0 = 2.95 m/s). Images show (i) ice lamella removing after droplet impact freezing is complete and (ii) the surface immediately after ice sliding from the surface.

(B) Time-lapse top-view optical images of self-splitting of ice on the hydrophobic Cu surface (see Video S4) with ΔT_s = 75°C, D_0 = 3.7 ± 0.05 mm, T_d = 23 ± 0.5°C, and We = 383 (U_0 = 2.73 m/s).

(C) Optical image of ice self-peeling driven by sequential droplet impacts on a bare Cu surface with ΔT_s = 75°C, D_0 = 3.7 ± 0.05 mm, T_d = 23°C ± 0.5°C, and We = 308 (U_0 = 2.45 m/s). Numbers 1–4 denote the sequence of water droplet impacts.

ensure delayed thermal homogenization, inducing smaller thermal stresses within the ice. This result indicates that the low adhesion and rapid cooling are needed in conjunction to ensure passive self-deicing. Although the majority of conventional icephobic coatings have low thermal conductivity (polymers), low-adhesion coatings with high thermal conductivity are preferred for self-deicing in high supercooling conditions.[41,42]

In summary, our work demonstrates fast-freezing and thermal-contraction-governed low adhesion and peeling of ice formed during droplet impact icing on supercooled surfaces. The evanescent interactions between the instant freezing at the droplet-substrate interface and trapped air by impacting droplets accounts for the observed low adhesion between the ice and substrate. The thermal stress within the completely or partially frozen liquid film induces deformation (bending) of the ice lamella and spontaneously detaches the ice from the substrate. The thermal stress is prone to bending the ice lamella rather than generating cracks to release the thermal contraction energy when ice adheres poorly to the substrate. Hence hydrophobic coatings can trigger self-peeling at substrate supercooling of 29°C, which was 14°C lower than that on bare hydrophilic bare Cu surfaces. As a result, self-peeling further reduces the ice-substrate contact area, resulting in the apparent shear adhesion strengths as low as 0 kPa. Strong adhesion region is observed to exist at the central area of the impact frozen droplet. Employing thermo-mechanical stresses, self-deicing was realized by either peeling or bursting through manipulation of the droplet inertia (We), substrate supercooling ΔT_s, and surface wettability. Sequential overlapping droplets are shown to generate a directional thermal contraction stress to achieve passive deicing. The present study not only unveils the underlying mechanisms of ultra-low adhesion and self-deicing during droplet impact icing but also promises a new way to control the adhesion of impact droplet solidification.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the lead contact, Nenad Miljkovic.

Materials availability

This study did not generate new unique materials.

Data and code availability

Details about the model implementation and the data used in this study can be found in this article and in the supplemental information. Requests for further information should be directed to the lead contact.

Surface fabrication and characterization

Mirror-finish Cu plates (99.9% purity, McMaster; see Figure S9A)[43] were used as the impact surfaces because of their large effusivity, which lowers the interfacial temperature. By doing so, ice peeling, crack pattern transformation, and low adhesion can be visualized more readily, and underlying mechanisms can be revealed more clearly.

To make the hydrophobic Cu surface, we first ultrasonically cleaned mirror-finish Cu plates with acetone for 5 min and rinsed with ethanol, isopropanol, and deionized (DI) water and dried with nitrogen. To remove the native oxide, we then immersed the samples in a hydrochloric acid solution (HCl, 2 M) for 30 s and then rinsed with DI water and dried with a clean nitrogen stream. To functionalize the surface, we deposited heptadecafluorodecyltrimethoxysilane (HTMS) using vapor-phase deposition. The surface was placed in a beaker with a vial of 1 mL HTMS (CAS #: 83048-65-1; TCI America) toluene solution (5% v/v). A lid was placed on top to seal the container, followed by heating in an atmospheric pressure oven (Lindberg Blue M; Thermo Scientific) at 90°C $\pm$ 5°C for 3 h.

The CuO nanoblade surface (see Figure S9B) was fabricated with a well-established chemical-oxidation method.[44] The cleaned samples (the same cleaning method as described to create the hydrophobic Cu) were immersed into a hot (96°C ± 3°C) alkaline solution composed of $NaClO_2$, NaOH, $Na_3PO_4 \cdot 12H_2O$, and DI water (3.75:5:10:100 wt %). The oxidation process lasted for 10 min, after which the samples were removed from the solution, rinsed immediately with DI water, and dried with a clean nitrogen stream. To make the CuO surfaces superhydrophobic, we deposited a low-surface-energy HTMS (CAS #: 83048-65-1; TCI America) coating using vapor-phase deposition as described for the hydrophobic Cu surface. The SLIPS was prepared by infusing the superhydrophobic CuO surface with medium viscosity fluorinated Krytox oil: Krytox 1525 (Chemours).[45] Scanning electron microscopy (SEM) for CuO nanoblade surface was performed on an SEM (FEI Quanta 200 FEG) at an imaging voltage of 15 kV.

To measure the apparent contact angle, monodisperse water droplets with diameters ranging from 30 to 40 μm were dispensed by a piezoelectric micro-goniometer (MCA-3; Kyowa Interface Science) operating at 7 V and 30–60 Hz. The droplet on the surface was grown to ≈100 nL by accumulating the dispensed droplets, and the advancing of the droplet contact line was recorded at 1 frame per second (fps) by an integrated camera and custom optics. The receding contact angle was measured by stopping dispensing droplets and letting the droplet shrink via evaporation. The advancing and receding contact angles were extracted by commercial software (FAMAS Interface Measurements & Analysis System; Kyowa) via the ellipse fitting method. Measurements were repeated on three to five different locations on the surface to obtain the averaged advancing or receding contact angles. The specific values of advancing and receding contact angles are listed in Table S2. Greater details of the contact angle measurement method can be found in past studies.[46]

Visualization of droplet impact icing

A schematic diagram of the test setup is shown in Figure S10. A chiller (PolyScience IP-100) was used to cool down the cold stage where the test sample was located. The cold stage was insulated in a transparent box made of acrylic. Before testing, nitrogen gas was charged into the test chamber at a rate of 1.5 L/min for at least 1 h to lower the environmental relative humidity (no more than 10% during the testing), inhibiting condensation and desublimation of water vapor on the test surfaces. When performing the test, a sample was first placed on the cold stage, and the T_s was measured by a sheathed thermocouple (T type; Omega) inserted 1 mm beneath the substrate top surface. To obtain good thermal and mechanical contact between the cold plate and sample, we fixed the sample on the cold stage by two screws at the edge. Next, DI water was pumped by a syringe pump (NE-1000; NewEra) at a very low flow rate of 0.05 mL/min, making inertial effects negligible, and dispensed from a needle (Terumo). The droplet size was controlled by selecting the diameter of the needle used in the experiment. The droplets fall because of gravity and impact onto a horizontally oriented supercooled surface. The main parameters, which characterize impact freezing, include the degree of substrate supercooling $\Delta T_s = T_m - T_s$, droplet initial temperature T_d, droplet Weber number, $We = \rho D_0 U_0^2/\sigma$, where $T_m = 0°C$ is the freezing temperature of water at normal atmosphere, T_s is the substrate temperature, and ρ and σ denote the density and surface tension of the liquid water, respectively, at $T_d = 23°C \pm 0.5°C$. Before a droplet collides with the solid surface, its shape can be approximated as an ellipse. The equivalent diameter of the droplet is calculated by $D_0 = (D_v D_h^2)^{1/3}$, where D_v and D_h are the vertical and horizontal lengths of the droplet, respectively. D_0 is

fixed at 2 ± 0.05 mm in all tests unless specified. Given the relatively low droplet impact velocity with negligible air drag, the impact velocity of droplets (U_0) can be estimated by $U_0 = \sqrt{2gH}$, where g = 9.81 m/s^2 is the gravitational acceleration and H is the falling height of the impacting droplet. Two high-speed cameras, a Phantom v711 (Vision Research) and a FASTCAM Mini AX200 (Photron), were mounted orthogonally on two movable stages to acquire images from the side and top, respectively. The measurement uncertainty of peeling height is estimated to be <14%.

Apparent shear adhesion strength measurement

In contrast with conventional adhesion tests used for static ice cubes,[47] our *in situ* adhesion characterization method enables the characterization of ice formation during dynamic droplet impact. However, to measure the impact icing adhesion strength, we constructed an apparatus including a force gauge (DFS10; Nextech) and a linear stage (see Figure S11A) based on the above-described setup (see Figure S10) for visualization. The ice column formed by sequential droplets usually shows a cylinder shape with a hemispheric cap (see Figure S11B). During the test, the force gauge tip was driven parallel to the ice-surface interface toward the ice columns at a low speed of 0.1 mm/s until the ice column detached from the surface. The probe was located close to the tested surface (less than 1 mm) to minimize the torque within the ice cylinder. The variation of force (see Figure S11C) was recorded by a computer, and the adhesion strength was calculated by dividing the peak force measured by the vertically projected area of the ice column. Because a single ice flake was too thin to measure its adhesion with the impacted surface directly, two ice soldering methods were attempted. Two purple columns in Figure S11D represent the single droplet soldering method where one droplet with very small falling height (H = 5 mm) was deposited on the ice flake, then the force gauge tip pushed the ice from the soldering droplet. The two blue columns in Figure S11D denote the ice soldering method used in our work. The two methods generally exhibit similar results despite the fact that a slightly higher value was acquired by the ice column (multiple droplets) method. Considering the ease of implementation, the ice column method was employed.

Note that the *in situ* ice adhesion measurement method used in this paper aims to characterize the low adhesion after self-peeling of ice flake, it is not applicable to measurements where the adhesion force is too large (≥10 N). When the adhesion force measured is too large, the tip of the force gauge (see Figure S11B) slips away from the non-flat vertical surface of the ice column. That is why few data of low-substrate supercooling (<35°C) are acquired; smaller supercooling means higher adhesion force, which is beyond the range of the present study.

Bending curvature calculation

The bending curvature (K) of the ice lamella was obtained by fitting the side-view bottom profile of the ice with a circular arc (see Figure S12), where L_e is the distance of two edges on the side-view image. Thus, K can be calculated as follows: $K = 8\delta/(4\delta^2 + L_e^2)$.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.xcrp.2022.100894.

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AUTHOR CONTRIBUTIONS

X.Y., P.J., and N.M. conceived the initial idea for this research. X.Y., W.T., and N.M. guided the work. P.J., X.Y., K.F.R., S.S., J.L., X.F., and S.C. fabricated and characterized the experimental samples. P.J., X.Y., M.J.H., K.F.R., S.S., J.M., J.C., and S.C. carried out the droplet impact icing experiments. P.J., X.Y., and J.M. analyzed the data and carried out the theoretical analysis. P.J. and X.Y. contributed equally to this work. All authors were responsible for writing the paper and have given approval to the final version of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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